

6 Monoclonal antibodies are used in many diagnostic tests.

Figure 6.1 shows stages in the production of a monoclonal antibody.

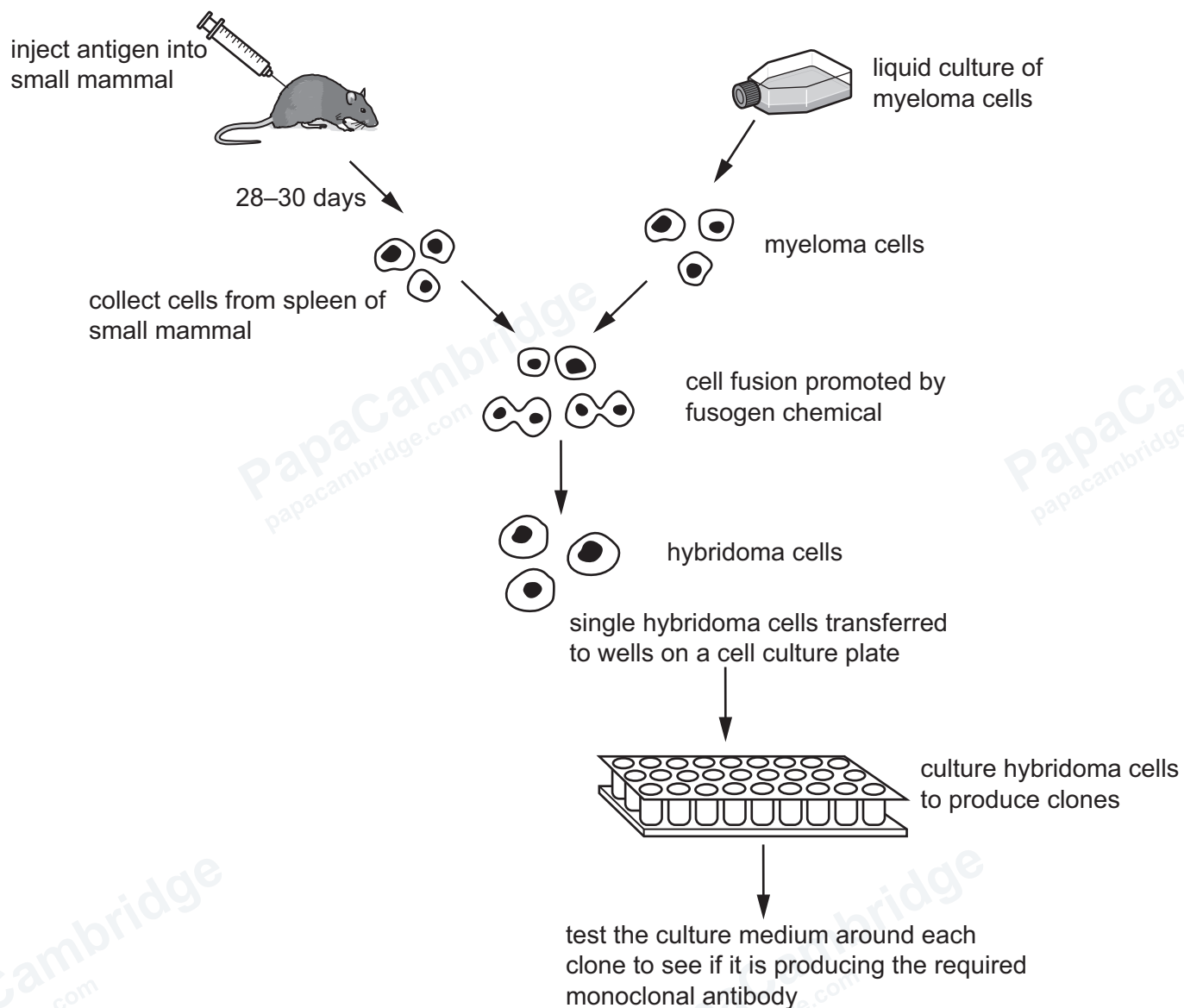


Figure 6.1

(a) State the name of the type of cell that is collected from the spleen of the small mammal.

..... [1]



- (b) Explain why myeloma cells are used in the production of monoclonal antibodies.

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..... [2]

- (c) Mpox is a viral disease. Scientists used the procedure shown in Figure 6.1 to develop a monoclonal antibody for use in diagnostic test strips.

The culture medium around each of the clones shown in Figure 6.1 was tested to find a suitable monoclonal antibody to detect the mpox virus in blood samples of people who might have been exposed to the disease.

Suggest how the culture medium around each of the clones can be tested to find a monoclonal antibody suitable for the diagnostic test strips.

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..... [1]

- (d) Monoclonal antibodies that are produced in the way shown in Figure 6.1 are not suitable for the long-term treatment of human diseases.

Scientists have solved this problem by producing monoclonal antibodies using small mammals in which the gene coding for the heavy polypeptides of their antibodies is replaced by the gene from humans.

Explain the advantage of using these modified small mammals in the production of monoclonal antibodies for use in the long-term treatment of human diseases.

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..... [2]

[Total: 6]